ML Lab-09 USN No.:CS23071 Aim:To implement the K-means clustering algorithm and analyze the clustering results using appropriate evaluation techniques.link text

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
path = '/content/drive/MyDrive/dataset_9/Mall_Customers.csv'
df = pd.read_csv(path)
df.head(5)
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Next steps: Generate code with df New interactive sheet

```
df.shape
```

```
(200, 5)
```

```
df.columns
```

```
Index(['CustomerID', 'Gender', 'Age', 'Annual Income (k$)',
       'Spending Score (1-100)'],
      dtype='object')
```

```
df.isnull().any().any()
```

```
np.False_
```

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

```
X.head()
```

	Annual Income (k$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

Next steps: Generate code with X New interactive sheet

```
X = pd.get_dummies(X,dtype = int)
```

```
X.head()
```

	Annual Income (k$)	Spending Score (1-100)
0	15	39
1	15	81
2	16	6
3	16	77
4	17	40

Next steps: Generate code with X New interactive sheet

```
x =df.iloc[:, [3,4]].values
print(X)
```

```
     Annual Income (k$)  Spending Score (1-100)
0                    15                      39
1                    15                      81
2                    16                       6
3                    16                      77
4                    17                      40
..                  ...                     ...
195                 120                      79
196                 126                      28
197                 126                      74
198                 137                      18
199                 137                      83

[200 rows x 2 columns]
```

```
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)
```

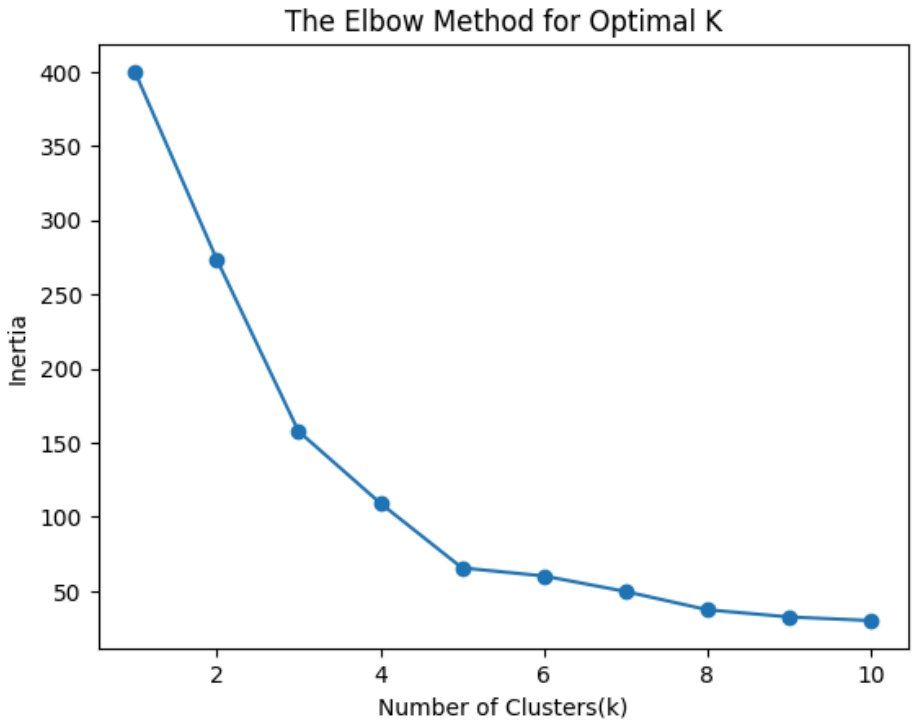
```
print(x_scaled)
```

```
[[-1.73899919 -0.43480148]
 [-1.73899919  1.19570407]
 [-1.70082976 -1.71591298]
 [-1.70082976  1.04041783]
 [-1.66266033 -0.39597992]
 [-1.66266033  1.00159627]
 [-1.62449091 -1.71591298]
```

```
 [-1.62449091  1.70038436]
 [-1.58632148 -1.83237767]
 [-1.58632148  0.84631002]
 [-1.58632148 -1.4053405 ]
 [-1.58632148  1.89449216]
 [-1.54815205 -1.36651894]
 [-1.54815205  1.04041783]
 [-1.54815205 -1.44416206]
 [-1.54815205  1.11806095]
 [-1.50998262 -0.59008772]
 [-1.50998262  0.61338066]
 [-1.43364376 -0.82301709]
 [-1.43364376  1.8556706 ]
 [-1.39547433 -0.59008772]
 [-1.39547433  0.88513158]
 [-1.3573049  -1.75473454]
 [-1.3573049   0.88513158]
 [-1.24279661 -1.4053405 ]
 [-1.24279661  1.23452563]
 [-1.24279661 -0.7065524 ]
 [-1.24279661  0.41927286]
 [-1.20462718 -0.74537397]
 [-1.20462718  1.42863343]
 [-1.16645776 -1.7935561 ]
 [-1.16645776  0.88513158]
 [-1.05194947 -1.7935561 ]
 [-1.05194947  1.62274124]
 [-1.05194947 -1.4053405 ]
 [-1.05194947  1.19570407]
 [-1.01378004 -1.28887582]
 [-1.01378004  0.88513158]
 [-0.89927175 -0.93948177]
 [-0.89927175  0.96277471]
 [-0.86110232 -0.59008772]
 [-0.86110232  1.62274124]
 [-0.82293289 -0.55126616]
 [-0.82293289  0.41927286]
 [-0.82293289 -0.86183865]
 [-0.82293289  0.5745591 ]
 [-0.78476346  0.18634349]
 [-0.78476346 -0.12422899]
 [-0.78476346 -0.3183368 ]
 [-0.78476346 -0.3183368 ]
 [-0.70842461  0.06987881]
 [-0.70842461  0.38045129]
 [-0.67025518  0.14752193]
 [-0.67025518  0.38045129]
 [-0.67025518 -0.20187212]
 [-0.67025518 -0.35715836]
 [-0.63208575 -0.00776431]
```

```
inertia_values = []
for k in range(1,11):
  kmeans = KMeans(n_clusters=k,random_state=42)
  kmeans.fit(x_scaled)
  inertia_values.append(kmeans.inertia_)
print(inertia_values)
plt.figure()
plt.plot(range(1,11),inertia_values,marker='o')
plt.xlabel('Number of Clusters(k)')
plt.ylabel('Inertia')
plt.title('The Elbow Method for Optimal K')
plt.show()
```

```
[399.99999999999994, 273.66888662642003, 157.70400815035939, 109.22822707921345, 65.56840815571681, 6
```




```
Kmeans = KMeans(n_clusters=9,random_state=42)
clusters = Kmeans.fit_predict(x_scaled)
df['clusters'] = clusters
print(df.head())
```

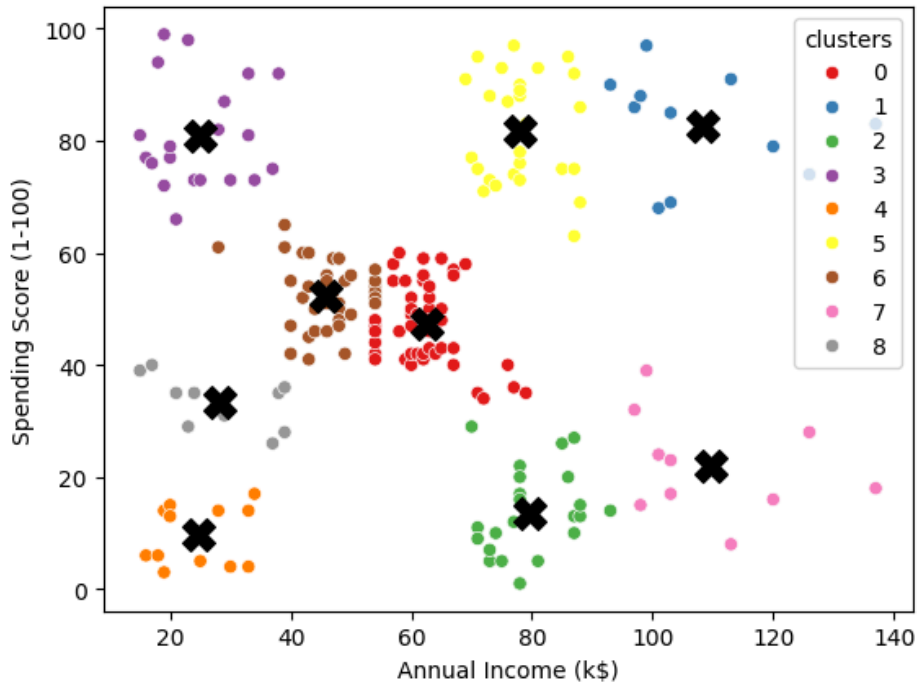
```
   CustomerID  Gender  ...  Spending Score (1-100)  clusters
0           1    Male  ...                      39         8
1           2    Male  ...                      81         3
2           3  Female  ...                       6         4
3           4  Female  ...                      77         3
4           5  Female  ...                      40         8

[5 rows x 6 columns]
```

```
print("Intertia:",Kmeans.inertia_)
sil_Score = silhouette_score(x_scaled,clusters)
print("Silhouette Score:",sil_Score)
```

```
Intertia: 32.495081199100916
Silhouette Score: 0.4566624374485964
```

```
plt.figure
sns.scatterplot(x='Annual Income (k$)',y='Spending Score (1-100)',hue='clusters',data=df,palette='Set
centroids = scaler.inverse_transform(Kmeans.cluster_centers_)
plt.scatter(centroids[:,0],centroids[:,1],marker='X',s=200,c='black',label='Centroids')
```

```
<matplotlib.collections.PathCollection at 0x78ded92b9250>
```



Start coding or generate with AI.